

A conjecture about direct product of funcoids

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Énoncé original (anglais, reproduction mot pour mot)

Conjecture Let f_1 and f_2 are monovalued, entirely defined funcoids with $\text{Src } f_1 = \text{Src } f_2 = A$. Then there exists a pointfree funcoid $f_1 \times^{\leftarrow D \rightarrow} f_2$ such that (for every filter x on A) $\left\langle f_1 \times^{\leftarrow D \rightarrow} f_2 \right\rangle x = \bigcup \left\{ \left\langle f_1 \right\rangle X \times^{\text{FCD}} \left\langle f_2 \right\rangle X \mid X \in \text{atoms} \left(\frac{A}{\times} \right) \right\}$. (The join operation is taken on the lattice of filters with reversed order.)

A positive solution of this problem may open a way to prove that some funcoids-related categories are cartesian closed.

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